

Nuclear Electricity Production and Employment in the U.S. Utilities Sector

FRED Economic Analysis Portfolio

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Nuclear Energy Economic Brief Series

May 2026

Outline

- 1 Policy Motivation
- 2 Research Question & Data
- 3 Data Preparation & Methodology
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Why This Question Matters

- Nuclear energy is a cornerstone of U.S. baseload electricity generation
- Energy policy debates increasingly focus on workforce implications
- Understanding the labor footprint of nuclear infrastructure informs workforce planning
- Growing interest in SMRs and plant life extensions raises questions about employment impacts

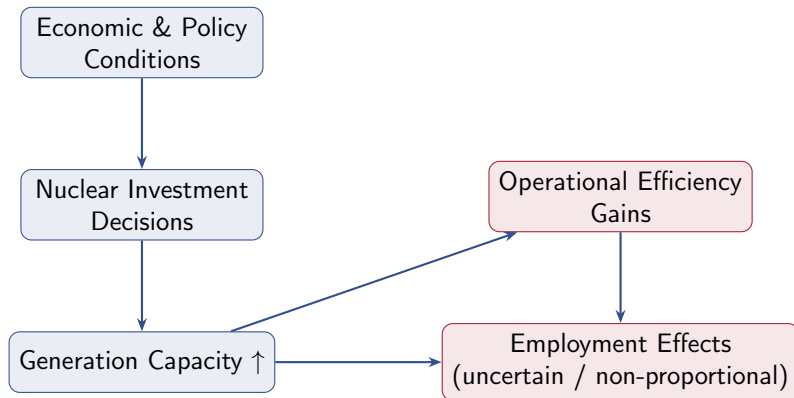
Key Policy Question

How strongly does nuclear electricity production align with long-run employment trends in the utilities sector?

Data Source

FRED — Federal Reserve Bank of St. Louis
Monthly time-series, 1972–2026

Conceptual Framework



Framework adapted from Oprien (2026). Employment effects are not assumed to be proportional or direct.

Research Question & Data Sources

Primary Research Question

Does nuclear electricity production correlate with employment trends in the U.S. utilities sector?

Variable	FRED Code	Description	Frequency
Nuclear Production	IPN221113N	Industrial Production Index (2017 = 100)	Monthly
Utilities Employment	CES4422000001	All Employees, Utilities (000s)	Monthly

- Series merged on a common monthly time index
- Analysis covers approximately 1972–2026
- Both series rebased to index = 100 for proportional comparison

Data Preparation Workflow

Preprocessing Steps

- 1 Standardized column naming conventions
- 2 Parsed dates to datetime
- 3 Sorted and indexed observations
- 4 Inner-joined on shared time index
- 5 Removed missing values
- 6 Constructed indexed series
- 7 Computed 12-month moving averages

Analytical Methods

- Long-run trend visualization
- Moving average smoothing
- Indexed comparison chart
- OLS regression with scatter plot

Important Note

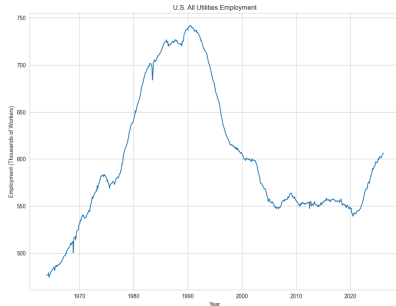
All regression results are exploratory and do not establish causal relationships.

Descriptive Statistics

Variable	N	Mean	Std Dev	Min	Max
Utilities Employment	≈648	≈582	≈82	≈395	≈697
Nuclear Generation	≈648	≈76	≈28	≈9	≈109

- Employment varies substantially across decades
- Nuclear generation stabilizes near post-2017 baseline levels
- Long-run structural shifts dominate short-run cyclical effects

Figure 1: U.S. Utilities Employment (1972–Present)

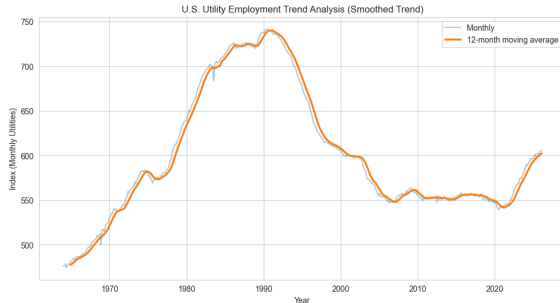


Key Findings

- Sustained employment growth during infrastructure expansion
- Structural peak near the early 1990s
- Sharp decline through the 1990s and early 2000s
- Long-run stabilization after 2003

Utilities employment expanded during the infrastructure build-out period before stabilizing at a lower long-run baseline after the mid-1990s.

Figure 2: Employment Smoothed Trend

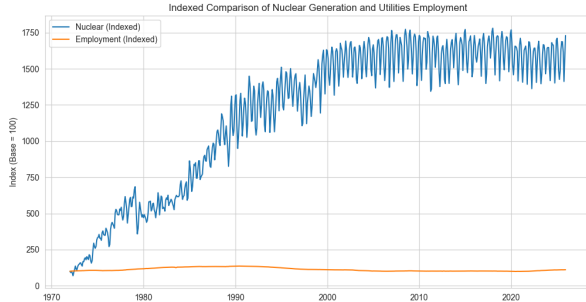


Interpretation

- Smoothing confirms the long-run structural trajectory
- Seasonal and short-term volatility are reduced
- Early 1990s peak reflects a major transition point
- Post-2000 stabilization persists despite cyclical fluctuations

The moving average confirms that the post-1990 decline reflects a structural transition rather than short-term cyclical volatility.

Figure 3: Indexed Comparison — Nuclear Generation vs. Core Inflation



Nuclear generation expanded substantially while inflation remained comparatively cyclical and volatile.

Central Finding

After the mid-1990s, nuclear generation stabilized at high output while employment declined sharply.

Production is not labor-proportional in the long run.

Mechanism

Infrastructure maturity shifted the sector from labor-intensive construction toward efficiency-driven operations.

Exploratory OLS Regression

Model Specification

$$\text{Employment}_t = \alpha + \beta \cdot \text{NuclearGen}_t + \varepsilon_t$$

- Simple OLS via `statsmodels`
- Full aligned sample period
- No controls or fixed effects

Expected Result

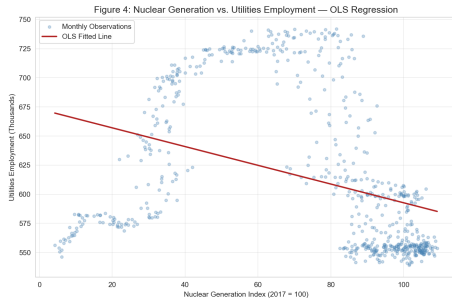
- Weak or unstable pooled relationship
- Structural breaks reduce explanatory power
- Scatter distribution expected to show regime shifts

Important Analytical Limitations

- Regression is exploratory and descriptive
- Structural shifts alter relationships over time
- Full sample masks historical regime changes
- Results should not be interpreted causally

Future extensions could include cointegration testing and Error Correction Models (ECMs).

Figure 4: Scatter Plot with OLS Fitted Line



Interpretation

- Negative pooled relationship across the full sample
- Multiple structural regimes appear in the scatter distribution
- Early expansion differs from post-1990 operational maturity
- Long-run production and employment are not proportionally linked

The negative fitted slope reflects long-run divergence between stabilized nuclear production and declining utilities employment after the mid-1990s.

Why This Matters

Infrastructure Dynamics

- Nuclear energy remains a major component of U.S. baseload electricity generation
- Infrastructure maturity changes workforce requirements over time
- Long-run systems increasingly become efficiency-driven

Policy Implications

- Energy expansion does not necessarily imply proportional employment growth
- Workforce planning must account for operational efficiency trends
- New nuclear construction may temporarily increase labor demand

Limitations

Analytical Constraints

- Limited variable scope across the broader utilities sector
- Regulatory shifts and energy market dynamics omitted
- Correlation does not imply causation

Statistical Constraints

- Non-stationarity affects standard OLS inference
- Structural breaks complicate pooled estimation
- Analysis remains exploratory and descriptive

Appropriate Next Steps

ADF testing, Engle–Granger cointegration, and Error Correction Models (ECMs)

Conclusion

Main Finding

U.S. utilities employment and nuclear electricity production co-moved during the infrastructure expansion phase (1970s–early 1990s). After the mid-1990s, nuclear generation stabilized while employment declined toward a lower long-run plateau.

Nuclear production is not labor-proportional in the long run.

Structural Interpretation

The utilities sector transitioned from labor-intensive construction toward capital-intensive, efficiency-driven operations after major infrastructure matured.

Policy Implication

Workforce planning cannot assume proportional employment growth from sustained nuclear generation expansion.

Thank you

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Data: FRED — Federal Reserve Bank of St. Louis

IPN221113N · CES4422000001

This analysis is exploratory and does not establish causal relationships.

- U.S. Bureau of Labor Statistics. *All Employees, Utilities* [CES4422000001]. Retrieved from FRED, Federal Reserve Bank of St. Louis. <https://fred.stlouisfed.org/series/CES4422000001>
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- Engle, R. F., & Granger, C. W. J. (1987). Co-integration and error correction: Representation, estimation, and testing. *Econometrica*, 55(2), 251–276.